Chapter 2 · Story

A brighter poster. More visitors. A cause?

An original story at the Common Ground repair café.



The poster that “worked” · c02s01

A fictional story. A real question about people.

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What would you do?

Which statement fits Leo’s evidence?

A The poster caused the increase.

B The poster had no effect.

C Attendance rose; several causes remain possible.

Suggested answer: C. Saying that attendance rose preserves the observation. Leaving the cause open reflects the limits of the comparison.

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Together does not establish because.

- **Correlation**: two things vary together.
- **Observed**: attendance rose after Leo changed the poster.
- **Causal claim**: the poster produced the rise.
- **Check**: what else could explain the change?

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Together does not establish because

Observed pattern



Poster changed.
Attendance rose.
Two things varied together.

Possible explanations



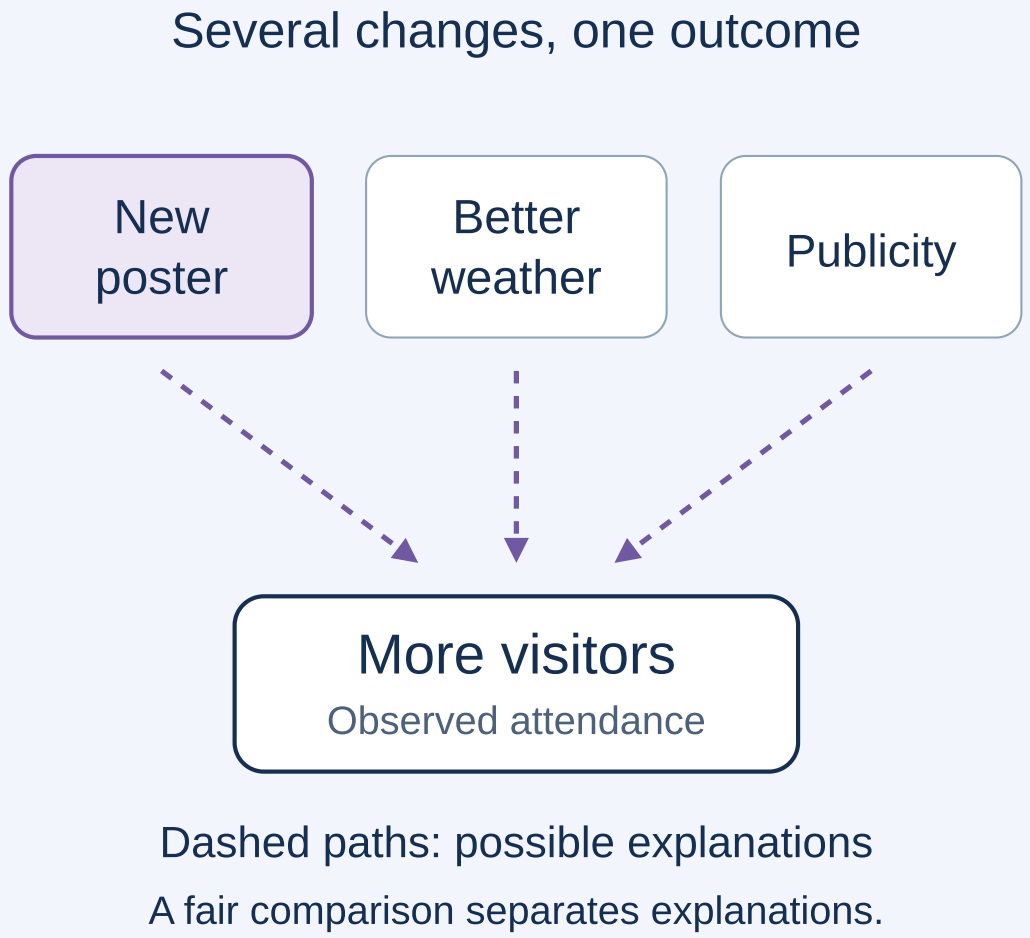
Poster?
Weather?
Publicity?
What would separate these explanations?

Correlation describes a relationship.
A causal claim needs a fairer test.

A shared pattern leaves multiple causes possible.

Several changes, one result.

- **Confound**: another factor could explain the pattern.
- **Poster, weather, publicity**: all changed together.
- **One outcome**: more people attended.
- **Problem**: this comparison cannot isolate the poster’s contribution.



Several changes can fit the same before-and-after pattern.

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Select people. Assign conditions.

- **Random sampling**: select people from a population by chance.
- **Random assignment**: allocate conditions by chance.
- **Same event**: compare invitation versions A and B.
- **Limit**: assignment does not make a sample representative.

A possible selection-and-allocation design

1. **Population**
   Residents we want to learn about
2. **Random sampling**
   Select people by chance
3. **Sample enters the comparison**
   Who was actually included?
4. **Random assignment**
   Allocate invitation A or B by chance

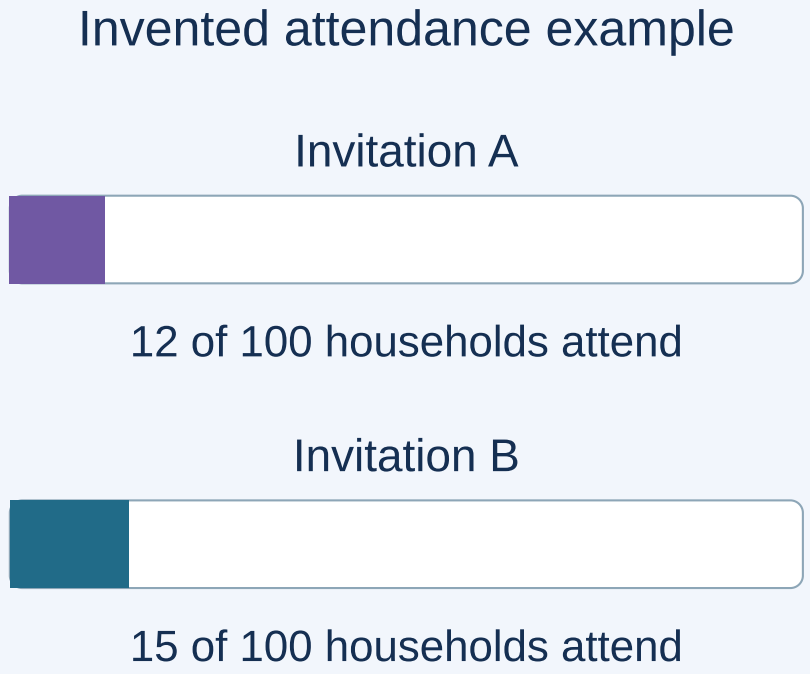
Assignment does not make a sample representative.

Choosing the sample and assigning conditions answer different questions.

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How large is the difference?

- **Invented example**: 12 of 100 attend versus 15 of 100.
- **Difference**: three percentage points.
- **Uncertainty**: another sample may give a different estimate.
- **Decision**: consider the size, precision, and cost.



Invented counts illustrate the size of a difference.

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Significance and usefulness are different questions.

- **Statistical significance**: evidence assessed against a specified model.
- **Practical importance**: is the difference useful for this decision?
- A significant result may be too small to matter.
- A nonsignificant result does not prove zero effect.

Two questions about a result

Statistical significance



How unusual under a specified model? Often a model of no group difference.

Practical importance



How large is the difference? Is it useful enough for the decision?

Significance does not measure usefulness.
Nonsignificance does not prove zero effect.

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Model-based evidence and practical value are different questions.

What can we really conclude?

Replication repeats a test to examine how dependable a finding is. Independent tests help us learn whether a result holds up elsewhere. Preregistration records the planned methods and analysis before results are known. That advance record reduces some opportunities for selective reporting. Preregistration cannot rescue a poor design. With a sound design, a larger sample can narrow the uncertainty around an estimate. More observations cannot fix a systematically unfair comparison. Sharing the study's methods and materials lets others inspect how it was done. A memorable study is a starting point for evaluation, rather than a final verdict.

Build confidence through checks

1. **Preregister**
   Record the plan before results
2. **Run a sound comparison**
   Sample size can improve precision
3. **Share methods and materials**
   Make the work inspectable
4. **Replicate independently**
   Check how dependable the finding is

A larger sample cannot fix a biased comparison.

Plan, inspect, and independently repeat a sound comparison.

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Try the idea somewhere new.

Which invitation comparison is strongest?

A Design A in winter; design B in summer.

B Randomize designs for one event; define attendance in advance.

C Design A for existing fans; design B for newcomers.

Suggested answer: B. Randomizing invitations for one event improves the comparison. Also check whether households share invitations.

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Close your notes. Explain it.

> How does random assignment differ from random sampling?

> Why does the rise in attendance after Leo changed the poster fail to establish that the poster caused the increase?

1. Random assignment allocates people to conditions. Random sampling selects people from a population.

2. Other conditions can change over time and produce the same before-and-after pattern.



Maya, Sam and Leo at the repair café.

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Carry one idea into your week.

Find an everyday causal claim. Name its outcome, one competing explanation, and a feasible comparison. State whose behavior the result would represent.

Revisit tomorrow and in a week.
Explain it without notes, then use a different example.



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Take the question into a different setting.

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